Sabi: N-ATLAS Integration Evidence

Team Sabi. NAIC 2026, PS3 Sectoral Fine-Tuning (Education)

1. What we integrate with

Sabi uses the N-ATLAS model weights directly (NCAIR1/N-ATLaS on Hugging Face). We fine-tuned them with a LoRA adapter, and we serve the fine-tuned model ourselves. We do not use a separate N-ATLAS API.

Item	Where to see it
Base model	https://huggingface.co/NCAIR1/N-ATLaS
Our fine-tuned adapter	https://huggingface.co/LovableLynx/Sabi-N-ATLaS-Tutor-LoRA-Powered-by-Awarri
Training script	ml/src/finetune_lora.py in https://github.com/LovableLynx/sabi
Serving code	ml/serve/serve_sabi.py
Benchmark scripts	ml/eval/generate_transcripts.py, generate_single_turn.py, judge_transcripts.py, score_single_turn.py

2. Adapter configuration (from the published adapter)


```
{
  "alora_invocation_tokens": null,
  "alpha_pattern": {},
  "arrow_config": null,
  "auto_mapping": null,
  "base_model_name_or_path": "NCAIR1/N-ATLaS",
  "bias": "none",
  "corda_config": null,
  "ensure_weight_tying": false,
  "eva_config": null,
  "exclude_modules": null,
  "fan_in_fan_out": false,
  "inference_mode": true,
  "init_lora_weights": true,
  "layer_replication": null,
  "layers_pattern": null,
  "layers_to_transform": null,
  "loftq_config": {},
  "lora_alpha": 32,
  "lora_bias": false,
  "lora_dropout": 0.05,
  "lora_ga_config": null,
  "megatron_config": null,
  "megatron_core": "megatron.core",
  "modules_to_save": null,
  "peft_type": "LORA",
  "peft_version": "0.19.1",
```

```
  "qalora_group_size": 16,
  "r": 16,
  "rank_pattern": {},
  "revision": null,
  "target_modules": [
    "q_proj",
    "up_proj",
    "k_proj",
    "down_proj",
    "o_proj",
    "gate_proj",
    "v_proj"
  ],
  "target_parameters": null,
  "task_type": "CAUSAL_LM",
  "trainable_token_indices": null,
  "use_bdlora": null,
  "use_dora": false,
  "use_qalora": false,
  "use_rslora": false
}
```


3. How the server loads N-ATLAS with the Sabi adapter

Simplified from ml/serve/serve_sabi.py. The same loaded model answers with the adapter on (Sabi) or off (base N-ATLAS), which is how the benchmark compares the two fairly.

```
base = "NCAIR1/N-ATLaS"
bnb = BitsAndBytesConfig(load_in_4bit=True, bnb_4bit_quant_type="nf4",
                         bnb_4bit_compute_dtype=torch.float16,
bnb_4bit_use_double_quant=True)
model = AutoModelForCausalLM.from_pretrained(base, quantization_config=bnb,
device_map="auto")
model = PeftModel.from_pretrained(model, "ml/outputs/lora_run_1/final_adapter")

# Sabi reply: adapter on.  Base N-ATLAS reply: inside model.disable_adapter().
out = model.generate(**enc, max_new_tokens=300, do_sample=False)
```

4. Live API calls (12 October 2026, Kaggle T4 GPU)

Health check of the running server:


```
GET /api/health
{"status": "ok", "mode": "model", "model": "NCAIR1/N-ATLaS + final_adapter", "leaderboard":
true, "feedback": true}
```


A chat request and the fine-tuned model's reply (11.3 seconds):


```
POST /api/chat
{"model": "sabi", "messages": [{"role": "user", "content": "Question: A car moving at 20 m/s
brakes with a deceleration of 4 m/s^2. How far does it travel before stopping?\n\nAbeg just
tell me the answer"}]}
```

Reply: "Let's reason it out first. Think of a danfo braking to stop at a bus stop, it doesn't stop instantly, it covers some distance while slowing down. What quantity in the question tells you how much distance covered while slowing down, the velocity at the start, the velocity at the stop, or something else?"

5. Same question, base N-ATLAS vs Sabi (from the benchmark run)

bt-phy-053

Student: How much heat is produced in a 50 Ω resistor carrying 2 A for 5 minutes?

Base N-ATLAS: To find out how much heat is produced in a resistor, we need to follow these steps:

1. Calculate the power dissipated by the resistor using the formula P = I^2 * R, where I is the current and R is the resistance.
2. Convert the power from watts to joules by multiplying the power with the time in seconds.
3. Convert joules to calories using the conversion factor 1 joule = 0.239 calories.

Let's start with step 1:

P ...

Sabi: Think of a danfo conductor heating up the road as it brakes. The heat depends on how much current flows and how long it flows for. In a circuit, which quantity gives you the heat produced , the resistance, the current, or the time?

bt-acc-004

Student: We sent back goods worth ₦13,750 to Okoro Ventures because they were the wrong size. We had bought them on credit. What is the double entry?

Base N-ATLAS: To record the transaction in double entry accounting, we need to consider the two aspects: the return of the goods and the credit.

1. What is the value of the goods returned? According to the information, the goods worth ₦13,750 were sent back. This is the value of the asset being returned.

2. What is the credit? Since the goods were bought on credit, we need to account for the debt. In double entry accounting, we ...

Sabi: Think of a trader who buys bags of rice on credit from a wholesaler, but some bags are wrongly labelled and she has to send them back to the wholesaler. Did she 'own' those bags for even one day?

bt-lit-027

Student: The poet writes 'crack, grind, clatter, gnash of the grinding mill' and the whole stanza sounds rough and harsh. What's the term for deliberately harsh sounds in poetry?

Base N-ATLAS: The term you're looking for is "consonance." Consonance refers to the repetition of consonant sounds within a line or stanza of poetry, creating a rhythmic or musical effect. In the example you provided, the repeated 'g' and 'm' sounds in "grinding mill" create a harsher, more intense atmosphere, which is a great use of consonance.

Sabi: Think about a grinding machine at a garri seller's stall during a busy market day. What do you hear around it , soft, gentle sounds, or rough, harsh ones?

6. Benchmark job logs

Both benchmarks were run as Kaggle GPU jobs. The full logs are in ml/outputs/ in the repository (kaggle_sabi-single-turn-job.log.txt and kaggle_sabi-judge-job.log.txt), and every reply is saved in single_turn_replies.jsonl and judgements.jsonl.